

Pierpasquale Colagrande

Computer Vision Data Scientist

Stockholm, Sweden

Experience

Computer Vision Data Scientist

Arkion • Full-time

Mar 2026 - Present
Stockholm, Sweden

R&D in computer vision to improve the company's electrical power grid component analysis platform

- Implemented **state-of-the-art CNN** and **transformer-based vision models** and applied **advanced deep learning techniques** with PyTorch
- Developed and deployed a **synthetic data generation pipeline** using **Blender 3D modeling** and **Modal**
- Maintained and developed an internal **CLI tool** for quick deployment and update of machine learning models
- Created, cleaned and maintained **datasets, trained, tested and deployed models** for image classification and object detection
- Achieved an average of **~90% accuracy (mAP 50-95)** on multiple **object detection models**
- Implemented **analytics SQL queries** for **datasets and machine learning models** by means of **Google BigQuery** and **Metabase**

Additional initiatives

- Implemented and deployed an **autoresearch AI agent** for **improving generation of synthetic data**
- Implemented and fine-tuned a **multimodal System One model** based on **Qwen VLMs**
- Developed **agent skills and MCP tools** for internal purposes
- Implemented **DevOps/MLOps pipelines** with **GitHub, Docker, AWS** and **Terraform**
- Instructed **annotation teams** on how to **build datasets** for image classification and object detection tasks
- Designed and conducted **technical interviews and assignments** for **data science roles** across multiple seniority levels

Artificial Intelligence Engineer

Blimp.ai • Full-time

Feb 2024 - Feb 2026
Milan, Italy

R&D for the company's flagship product, a computer vision sensor designed for urban monitoring and analysis

- Increased inference speed **32x**, from **3 to 97 FPS** by **optimizing the codebase** and **porting it to NVIDIA Jetson Orin Nano** with **DeepStream**
- Achieved a **63% increase in people-counting accuracy** by upgrading to **Ultralytics YOLOv11-Small + BoT-SORT** and **optimizing hyperparameters**
- Achieved **92% accuracy on staff member classification** by training image classification networks
- Created, cleaned, and prepared **datasets for image classification and object detection**
- Reduced **offline sensors by 65%** by designing and developing a **robust input module for reliable IP camera connectivity** using **GStreamer**
- Enabled a **single sensor to handle multiple cameras** without **FPS loss**, leveraging **NVIDIA DeepStream optimizations**

Additional initiatives

- Created and maintained an **updater tool** for seamless deployment of packages, patches and new features to sensors in production
- Created and maintained a **GUI-based tool** for remote sensor management
- Built **REST APIs** and features tailored to various clients

Data Scientist

Lutech • Full-time

May 2023 - Jan 2024
Cinisello Balsamo, Italy

AI R&D projects

- Fine-tuned **vision models (CNNs, ViTs)** for **detection and classification on MRI images & multimodal datasets**, using **Hugging Face**
- Designed and developed an intelligent hydraulic plant automation system leveraging **Reinforcement Learning and Physics-Informed Deep Learning**
- Created **algorithms to convert hydraulic plant schematics into Deep Neural Network (DNN) architectures** via modular building blocks

AI consulting projects

- Delivered **Large Language Models (LLMs) and Retrieval-Augmented Generation (RAG) solutions**, including chatbots and text document analysis
- Implemented vision systems with **Vertex AI and Google Cloud Platform**
- Built **CI/CD pipelines** using **Docker, Kubernetes, and Mia-Platform**

Master's Thesis Intern

Univrses • Internship

Mar 2022 - Jul 2022
Stockholm, Sweden

R&D of a copy-paste data augmentation method for generating synthetic object detection datasets, deployed in the company's training pipeline

Education

Artificial Intelligence

University of Bologna • Master's Degree

Sep 2019 - Feb 2023
Bologna, Italy

Topics: Machine & Deep Learning, Natural Language Processing, Computer Vision, Reinforcement Learning, Knowledge Engineering • GPA: 3.73/4

Computer Science

University of Bari • Bachelor's Degree

Sep 2015 - Apr 2019
Bari, Italy

Topics: Algorithms and Data Structures, Databases, Object-Oriented Programming, Software Engineering • GPA: 3.86/4

Publications

CRISPRLearner: A Deep Learning-Based System to Predict CRISPR/Cas9 sgRNA On-Target Cleavage Efficiency 

MDPI, *Electronics* • December 2019

Projects

MPRNet: CNN for Super Resolution 

Sep - Dec 2022

Deep Reinforcement Learning for Atari 2600 

Jul - Sep 2022

BERT Fine-tuning for Question Answering 

Jan - Apr 2021

Conditional GAN for Face Generation 

Apr - Jul 2020

More on **GitHub** 

Technical & Soft Skills

Artificial Intelligence: Image Classification, Object Detection & Tracking, Edge AI, CNNs, Transformers, VLMs, LLMs, Agents, RAG, Machine & Deep Learning, Reinforcement Learning, Knowledge Graphs

Languages: Python, Shell Scripting, SQL, HTML/CSS, PHP, JavaScript, C, C++, Java, Scala

Cloud & Databases: AWS, Modal, Google Cloud Platform, Vertex AI, BigQuery, Metabase, SLURM, Dremio

DevOps & MLOps: Git, GitHub, GitLab, Bitbucket, Docker, Kubernetes, GitHub Actions, Terraform, Weights & Biases, **MLflow**, TensorBoard

Frameworks & Libraries: PyTorch, TensorFlow, OpenCV, NVIDIA Jetson, GStreamer, DeepStream, TensorRT, ONNX, LangChain, crewAI, smolagents, Keras, OpenAI, OpenAI Gym, Scikit-learn, Hugging Face, Pandas, NumPy, FastAPI, pytest

Agile: Scrum, Kanban, Jira, Confluence, Notion

Soft Skills: Problem Solving, Leadership, Continuous Learning, Proactivity, Curiosity, Ambition, Teamwork, Communication, Attention to Detail